Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
5	(a)		Net / resultant / [vector] sum of / total force (1) Correct example showing net of two or more forces numerically, algebraically or by convincing diagram (1)	1	1		2		
	(b)	(i)	Weight of block = 14 715 [N] (1) $a = \frac{(16\,000 - 14\,715)}{1500}$ (1) substitution and re-arrangement a = 0.86 [m s^{-2}] (1) **Alternative:** $a = \frac{16\,000}{1500}$ = 10.67 [m s^{-2}] (1) a = 10.67 – 9.81 (1) a = 0.86 [m s^{-2}] (1)	1	1 1		3	3	
		(ii)	Depends on acceleration (1) may exceed 16 000 N, so no (1)			2	2		
	(c)		Stress = $2 \times 10^{11} \times 3.2 \times 10^{-5} = 6.4 \times 10^{6}$ [N m^{-2}] (1) Tension = $6.4 \times 10^{6} \times 2 \times 10^{-3}$ = 12 800 [N] (1) **Award 2 marks** for determining T directly using strain $\times E \times A$ Decelerating – value not required but may be seen (1) **ecf from T** Because T < 14 715 [N] (can be implied) (1) **Alternative for 3rd and 4th marks** ΣF = 12 800 – 14 700 = -1 900 [N] (1) Negative value indicates a negative acceleration so the load must be decelerating (1)		4		4	2	

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			Alternative: Stress with weight only = $\frac{14\,715}{2 \times 10^{-3}}$ = 7.36 M[Pa] (1) Strain for this force = $\frac{7.36 \times 10^{6}}{2 \times 10^{11}}$ = 3.7×10^{-5} (1) Comparing with 3.2×10^{-5} (in question and might be implied) (1) Hence decelerating (1)						
			Question 5 total	**2**	**7**	**2**	**11**	**5**	**0**